

Food Delivery Business Analytics Case Study

1. Business Context

The business operates in the food delivery space in the Delhi NCR region and serves customers through the Zomato platform. Its business activity depends on customers placing orders and restaurants successfully preparing and fulfilling them. The business also uses discounts to attract customers and encourage orders. As the number of orders grows, it becomes important to understand customer behaviour, restaurant performance, demand patterns, and operational issues to maintain a good customer experience and support overall business performance.

2. Business Problem

The business does not have a clear understanding of what is driving its overall performance. It needs to better understand customer behaviour, retention and value, operational performance, cancellations, and differences across restaurant partners. Without these insights, it is difficult to identify which areas actually need attention and where business improvements could have the greatest impact.

3. Business Objectives

The objective of this analysis is to use historical order data to :

- Understand customer ordering and demand patterns.
- Understand customer retention and customer value.
- Evaluate operational performance and identify important operational relationships.
- Examine cancellation patterns and investigate factors associated with them.

- Compare restaurant performance across different business and operational metrics.
- Use statistical analysis to distinguish meaningful patterns from differences that are only statistically detectable.
- Identify actionable findings and provide data-driven recommendations for business decision-making.

4. Data Understanding

The dataset contains 21,321 food delivery orders recorded between 1 September 2024 and 31 January 2025. It contains 29 columns covering different parts of the order journey, including restaurant information, customer details, pricing, operational metrics, cancellations, and customer feedback.

Each row represents one food delivery order, and **Order ID** is unique for every order.

Business entities

- The dataset contains 6 restaurant brands operating across different subzones in Delhi NCR.
- There are 21 unique Restaurant IDs, meaning the same restaurant brand can have different IDs across different subzones.
- The dataset contains 11,607 unique customers, showing that some customers placed multiple orders during the study period.

Main types of information available

The columns can broadly be grouped into:

- Restaurant information : Restaurant ID, Restaurant Name, Subzone, City
- Order information : Order ID, Order Placed At, Order Status, Distance, Items in Order
- Financial information : Bill Subtotal, Packaging Charges, Discounts, Total Order Amount

- Operational information : Kitchen Preparation Time (KPT), Rider Wait Time, Order Ready Marked
- Customer information : Customer ID, Rating, Review, Complaint Tag

Important observations from the initial inspection -

A few columns required additional preparation before analysis :

- `Order Placed At` was stored as text and needed to be converted into a datetime format.
- `Distance` was stored as text values such as `<1km` and `3km`, so it needed to be converted into a usable numerical format.
- `Instructions` and `Review` were free-text fields and were not suitable for direct quantitative analysis without additional text processing.
- `Delivery` contained only one value, `Zomato Delivery`, so it did not provide useful variation for analysis.

Understanding missing values -

Missing values were not automatically treated as data-quality errors. Some fields are naturally missing because they are only applicable in certain situations.

For example :

- Ratings and reviews are available only when customers provide feedback.
- Complaint-related fields are relevant only when a complaint occurs.
- Cancellation-related fields are relevant only for cancelled or otherwise applicable orders.

Therefore, missing values were assessed based on the meaning and applicability of each field, rather than simply removing every row or filling every missing value.

5. Data Cleaning & Validation

After understanding the dataset, the next step was to check whether the available columns were useful and whether the data was suitable for analysis. The goal was not to remove data simply because it looked unusual, but to remove fields that provided no useful information and handle data based on its actual meaning.

Observation	Action Taken
Delivery contained only one unique value (Zomato Delivery).	Dropped because it had no analytical value.
City contained only one unique value (Delhi).	Dropped because it had no variation.
Restaurant penalty (Rejection) had only 3 non-null values, all equal to 0.	Dropped because it provided no useful information.
Instructions had around 97% missing values and contained unstructured text.	Dropped because there was insufficient usable information for quantitative analysis.
Review had around 98% missing values and contained personalized text.	Dropped because of very low coverage and unstructured content.
Customer complaint tag had around 97% missing values.	Dropped because there was insufficient coverage for reliable analysis.
Distance was stored as text values such as 2km and <1km .	Converted into a numerical format for analysis.
Order Placed At contained both date and time.	Extracted separate Order Date and Order Time fields and removed the original column.
Operational metrics such as KPT and Rider Wait Time contained some missing values.	Missing values were retained rather than artificially imputed, as they were treated as operational logging limitations.
Discount construct had complex and inconsistent missing-value patterns.	Retained without imputation or transformation at this stage.
Numerical variables such as Bill Subtotal showed right-skewed distributions and statistical outliers.	Values were retained because they appeared to be plausible business observations rather than clear data-entry errors.

Validation Approach -

The cleaned data was then reviewed to ensure that the transformations did not introduce obvious inconsistencies. Particular attention was given to unique IDs, data types, missing values, categorical values, and the ranges and distributions of important numerical variables.

An important principle throughout the cleaning process was that an unusual value was not automatically considered an error. For example, a highly skewed order value or a large rider wait time could represent a genuine business observation. Such values were therefore retained unless there was evidence that they were invalid.

This gave us a dataset that was suitable for the subsequent initial analysis, statistical validation, and driver analysis.

6. Business Questions

The analysis did not begin with a fixed list of questions. Instead, the questions were developed iteratively while exploring the data.

We first examined the available variables, distributions, customer behaviour, operational metrics, cancellations, and restaurant-level patterns. As interesting patterns or potential business issues emerged, they were converted into specific business questions for further investigation.

The questions were then grouped into four main areas:

- Demand & Temporal Patterns — How does order demand vary across time periods, days, and months ?
- Customer Retention & Value — How frequently do customers return, what drives customer value, and how is revenue distributed across customers ?
- Operational Performance & Cancellations — Which operational factors are associated with rider waiting, order-ready marking, and cancellations ?
- Restaurant Performance — How do restaurant partners differ in terms of demand, revenue, operations, pricing, and customer behaviour ?

Where an observed pattern had potential business importance, statistical testing was used to determine whether the difference or association was statistically detectable and whether it was large enough to have practical significance.

This iterative approach allowed the analysis to move from exploration → business question → statistical validation → deeper investigation, rather than forcing the data into a pre-defined set of questions.

7. Initial Analysis — Key Findings

The initial analysis was used to understand the main patterns in customer behaviour, demand, cancellations, operations, and restaurant performance before moving into deeper driver analysis.

Demand was strongly concentrated around certain periods. Evening / Dinner was by far the busiest time of day, while Late Night had much lower demand. Demand also varied across days of the week, with Saturday recording the highest order volume and Monday the lowest. In contrast, monthly demand remained relatively stable throughout the five-month study period.

Customer retention was relatively weak. Most customers placed only one order during the study period, while repeat customers represented a smaller share of the customer base. Retention also declined as more weeks passed after a customer's first order, with only a small proportion of customers continuing to order several weeks later.

Customer value was concentrated among repeat and high-spending customers. The top 20% of customers by spending contributed more than half of total revenue. Repeat customers represented only about one-third of customers but generated nearly two-thirds of revenue. Higher-value customers also tended to order more frequently and place higher-value orders.

Cancellation levels were relatively low overall. Both customer-initiated and platform-initiated cancellations represented a small proportion of total orders. Some differences were visible across distance groups, time periods, and other operational conditions, but these patterns required further investigation before being treated as meaningful business drivers.

Operational performance varied across different conditions. Rider wait time differed noticeably across order-ready marking statuses, with incorrectly marked orders showing substantially higher waiting times. Kitchen preparation time and rider wait time also showed some variation across restaurants and time periods, providing areas for deeper investigation.

Restaurant performance was highly concentrated. Aura Pizzas and Swaad together accounted for almost all platform orders and revenue. The restaurants also differed in preparation time, rider wait time, discounts, packaging charges, and order value, suggesting that restaurant performance was not uniform across partners.

These initial findings helped determine where deeper analysis was useful. Patterns that appeared potentially important were taken forward into statistical validation and driver analysis, while patterns with little practical difference were not treated as major business drivers.

8. Driver Analysis — Key Findings

After the initial analysis, the investigation focused on the patterns that appeared most relevant to the business. Statistical testing was used to separate differences that were statistically detectable from those that were large enough to matter in practice.

1. Operational Performance & Cancellation

The analysis did not identify a strong standalone factor explaining cancellation behaviour. Some variables, such as KPT and time of day, showed statistical relationships with cancellations, but their practical effects were small.

A clearer operational relationship was found between order-ready marking and rider wait time. Orders with incorrect or missed ready markings had noticeably higher rider wait times. However, the analysis of restaurant, KPT, and time-of-day patterns did not identify a strong factor explaining why problematic markings occurred.

Key takeaway : Cancellation does not appear to have one clear operational driver in the available data, while order-ready marking is an important operational issue because of its relationship with rider waiting.

2. Demand & Temporal Patterns

The strongest demand pattern was time of day. Evening / Dinner accounted for roughly 58% of orders and was clearly the busiest period.

Demand also varied across days of the week, particularly between Saturday and Monday, but the overall practical effect was much smaller. Monthly demand was even more stable, with only a negligible practical difference across the five months.

The Evening / Dinner peak was also visible across the major restaurants, and order values were generally higher during this period. Discounts, however, did not show a consistent increase during peak demand.

Key takeaway : Time of day is the strongest temporal demand pattern, while day-of-week and monthly differences are secondary.

3. Customer Retention

Customer retention declined as more time passed after the first order, falling from 8.90% in Week 1 to 4.66% by Week 10. Retention also differed across acquisition cohorts, with later cohorts generally showing weaker retention.

We then investigated whether operational experience, discounts, order value, and acquisition conditions could explain this decline. These factors did not show a strong or consistent relationship with the cohort-level retention pattern.

Key takeaway : Retention is a clear business concern, but the available data does not clearly explain why customers stop returning or why later cohorts retain less well.

4. Customer Value

Customer spending was highly uneven, with a relatively small group of high-spending customers contributing a large share of revenue.

Repeat customers represented only about one-third of the customer base but generated 62.96% of revenue, while the top 20% of customers by spending generated 53.50% of total revenue.

High-value customers were different mainly because they ordered more frequently and also had higher average order values. Order frequency showed the stronger difference. They also tended to reorder somewhat sooner, although the practical difference was small.

Key takeaway : Repeat purchasing is strongly linked with customer value, with order frequency being the strongest behavioural differentiator identified.

5. Restaurant Performance

Restaurant performance was highly concentrated. Aura Pizzas and Swaad together accounted for 97.93% of orders and 98.23% of revenue, making them the most important restaurant partners in the dataset.

Restaurants also differed in their operational and commercial characteristics, including KPT, rider wait time, discounts, packaging charges, and order value. Evening/Dinner was the dominant demand period for most restaurants.

Because the smaller restaurants had very low order volumes, their unusually high or low rates were treated as signals rather than reliable performance rankings.

Key takeaway : The platform's restaurant business is dominated by two partners, while restaurant-level operating and pricing patterns vary considerably.

9. Driver Analysis Summary

Overall Driver Analysis Takeaway :

The deeper investigation helped distinguish between patterns that looked interesting in the initial analysis and patterns that were actually meaningful from a business perspective.

The strongest findings were -

- Evening/Dinner demand is the clearest temporal pattern.
- Incorrect order-ready marking is associated with substantially higher rider wait time.
- Repeat customers and high-spending customers are disproportionately important to revenue.
- Customer retention is weak and declines across later weeks and cohorts, but its underlying drivers remain unclear.
- No strong standalone driver of Zomato cancellations was identified from the variables available.
- Restaurant performance is highly concentrated in Aura Pizzas and Swaad.

10. Recommendations and Business Impact

The analysis identified several areas that may be important for the business. These recommendations are based on the patterns and relationships observed in the data. Since this is observational data, the findings show associations and patterns rather than confirmed cause-and-effect relationships. Therefore, the recommendations should be tested before being implemented at scale.

1. Improve Customer Retention

What We Found

Customer retention declined from 8.90% in Week 1 to 4.66% by Week 10. Later acquisition cohorts also generally showed lower retention than earlier cohorts.

Further analysis examined KPT, rider wait time, discount, order value, and first-order conditions as possible explanations. However, these factors did not show a strong or consistent relationship with the decline in retention.

Why It Matters

Repeat customers are important for the platform because they generated 62.96% of total revenue while representing only 33.55% of customers.

Therefore, understanding and improving customer retention could have a meaningful impact on long-term customer value.

Possible Recommendation

The business should first investigate why customers are not returning rather than assuming that a particular operational or promotional factor is responsible.

Additional information such as customer feedback, acquisition source, restaurant preferences, and customer experience could help identify the reasons behind the retention decline.

Once a specific problem is identified, targeted retention strategies can be tested.

How Should the Business Test It?

The business can run controlled experiments where eligible customers are divided into:

- Control group: existing customer experience
- Treatment group: proposed retention intervention

This would help determine whether the intervention itself improves retention rather than simply showing that customers who received the intervention happened to return more often.

What Should Be Monitored?

- 28-day repeat-order rate
- Time to next successfully delivered order
- Second-order conversion
- Orders per customer
- Revenue/contribution per customer
- Cost of the retention intervention

When Should It Be Scaled?

The intervention should be expanded only if it produces a consistent improvement in retention and generates enough additional customer value to justify its cost.

2. Improve Order-Ready Marking

What We Found

Incorrect and missed order-ready markings were associated with substantially higher rider wait times.

This was one of the clearer operational relationships identified in the analysis.

Why It Matters

Longer rider waiting can reduce delivery efficiency and may create a poorer operational experience for riders and restaurants.

Possible Recommendation

The business could improve the order-ready marking process through:

- clearer guidance on when an order should be marked ready;
- restaurant staff training;
- monitoring restaurants with higher levels of incorrect or missed markings;
- regular review of rider wait time.

The objective should be to improve accuracy and timing of order-ready marking, rather than simply asking restaurants to mark orders ready earlier.

How Should the Business Test It?

A pilot can be conducted with selected restaurants where the improved process is introduced, while similar restaurants continue with the existing process.

The change in rider wait time can then be compared between the two groups.

What Should Be Monitored?

- Median rider wait time
- Order-ready marking accuracy
- Percentage of incorrect/missed markings
- Customer cancellation rate
- Zomato cancellation rate

When Should It Be Scaled?

The process should be expanded if the pilot shows a consistent reduction in rider wait time without creating inaccurate early order-ready marking.

3. Focus on High-Value and Repeat Customers

What We Found

The top 20% of customers generated 53.50% of total revenue.

High-value customers also:

- ordered more frequently;
- had higher average order values;
- tended to reorder somewhat sooner.

Among these factors, order frequency showed the strongest difference between high-value and other customers.

Why It Matters

A relatively small group of customers contributes a large share of platform revenue. Retaining these customers and encouraging sustainable repeat usage could therefore be important for maintaining customer value.

However, the analysis does not prove that simply increasing order frequency or reducing reorder time will cause customer value to increase.

Possible Recommendation

The business could:

- identify and retain existing high-value customers;
- identify promising repeat customers who may develop into high-value customers;
- test personalized engagement, recommendations, loyalty benefits, or targeted incentives.

The focus should be on sustainable repeat usage, rather than giving discounts to all customers.

How Should the Business Test It?

For selected customer groups, compare:

- Control: existing customer experience
- Treatment: proposed engagement or retention strategy

The business can then measure whether the treatment actually improves customer behaviour.

What Should Be Monitored?

- Customer retention
- Repeat-order rate
- Orders per customer
- Customer spending
- Average order value
- Movement into higher-value customer segments
- Intervention cost
- Revenue/contribution per customer

When Should It Be Scaled?

The strategy should be expanded when the improvement in customer behaviour is large enough to generate additional value that justifies the cost of the intervention.

4. Prepare Better for Evening/Dinner Demand

What We Found

Evening/Dinner was the strongest demand period, accounting for around 58% of all orders.

This pattern was also broadly visible across the major restaurant partners. For most restaurants, Evening/Dinner also had higher typical order values.

Why It Matters

A large proportion of daily demand is concentrated during this period. This makes peak-period operational readiness important for maintaining service efficiency.

However, the analysis does not show that Evening/Dinner demand itself causes higher cancellations or operational problems.

Possible Recommendation

The business could review peak-period:

- restaurant staffing;

- kitchen capacity;
- rider availability;
- order-ready marking;
- restaurants where operational performance becomes weaker during peak hours.

How Should the Business Test It?

For restaurants or areas showing signs of peak-period pressure, the business could run a targeted pilot involving additional staffing, rider capacity, or process improvements.

Performance can then be compared with similar restaurants or periods without the intervention.

What Should Be Monitored?

- Median KPT
- Median rider wait time
- Order-ready marking performance
- Cancellation rates
- Successfully fulfilled orders
- Cost of additional operational support

When Should It Be Scaled?

Peak-period changes should be expanded if they produce a meaningful improvement in operational performance relative to their cost.

5. Reduce Dependence on a Small Number of Restaurant Partners

What We Found

Restaurant performance was highly concentrated.

Aura Pizzas and Swaad together generated 97.93% of orders and 98.23% of revenue.

Why It Matters

This means that the platform's overall business performance is strongly dependent on a small number of restaurant partners.

A major operational or commercial problem affecting one of these restaurants could therefore have a relatively large impact on the platform.

Possible Recommendation

The business should continue maintaining strong relationships with the major restaurant partners while also developing additional restaurant partners that can generate meaningful demand.

The focus should not simply be on adding more restaurants, but on developing restaurants that can attract customers and generate repeat business.

How Should the Business Test It?

Selected smaller or new restaurants could receive additional support, visibility, or onboarding assistance.

Their performance can then be compared with similar restaurants that continue under the normal approach.

What Should Be Monitored?

- Order share from smaller/new restaurants
- Revenue contribution
- Unique customers
- Repeat customer rate
- Orders per restaurant
- Revenue per restaurant
- Overall restaurant concentration

When Should It Be Scaled?

The strategy should be expanded when additional restaurants demonstrate sustained customer demand, repeat usage, and meaningful revenue contribution.

Findings to Monitor Rather Than Act On Immediately

Not every statistically significant result requires an immediate business intervention. Several findings had very small practical effects or insufficient evidence to identify a clear driver.

- Late-night Zomato cancellations: cancellation rate was higher, but the practical effect was negligible. Monitor whether this pattern continues.
- KPT and Zomato cancellations: statistically significant difference was observed, but the effect was small and actually went in the opposite direction from what would be expected if longer KPT were driving cancellations.
- Day-of-week demand: Saturday and Monday showed the largest differences, but the overall practical effect was small. Use this mainly for routine planning.
- Monthly demand: November and January showed some differences, but the overall practical effect was negligible. More data is needed before treating this as a seasonal pattern.
- Customer frequency and discounts: statistically significant differences existed, but the practical effect was negligible. No major discount strategy change is supported by this finding.
- Reorder interval: high-value customers reordered somewhat sooner, but the practical difference was small. This should not be treated as a proven lever for increasing customer value.
- Small restaurant differences: some smaller restaurants showed extreme rates, but their low order volumes make these figures unstable.

Overall Business Impact

Overall, the analysis points to four main areas of business attention:

1. Customer retention — understand why customers stop returning and test targeted retention strategies.

2. Operational efficiency — improve order-ready marking because it has a clearer relationship with rider waiting.
3. Customer value — protect high-value customers and encourage sustainable repeat usage.
4. Restaurant operations and concentration — prepare for peak demand while gradually building a broader restaurant base.

The important point is that these are areas for action and testing, not claims of proven causation. The historical analysis tells us where the business should look more closely; controlled experiments and additional data are needed to determine which interventions actually improve business outcomes.

12. Limitations & Further Investigation

The analysis provides useful patterns and relationships in the available data, but there are some limitations that should be considered before making business decisions.

Limitations

- **Observational data:** The analysis identifies associations and patterns, but it does not prove that one factor directly causes another.
- **Limited study period :** The dataset covers only September 2024 to January 2025, so longer-term seasonal patterns cannot be confirmed.
- **Low cancellation volume :** Only a small number of orders were cancelled, especially for Zomato cancellations. This limits how confidently cancellation drivers can be identified.
- **Missing information :** Some variables, such as ratings, reviews, complaints, and cancellation-related information, were available only for a subset of orders.
- **Small restaurant samples :** Some restaurants had very few orders, so their percentages and rates may be unstable and should not be used for direct performance ranking.

- Limited customer information : The dataset does not contain information such as customer acquisition source, customer preferences, or detailed customer feedback. This limits our ability to explain why customers return or stop ordering.
- Limited operational factors : The available operational variables do not capture every factor that may affect cancellations, retention, or customer experience.
- Statistical significance vs. business importance : Some results were statistically significant but had very small effect sizes. These findings should therefore not automatically be treated as important business drivers.

Further Investigation

The analysis also highlights areas where additional data or deeper investigation would be useful.

1. Customer Retention

Retention declined across acquisition cohorts, but the factors examined did not clearly explain why. Further analysis could include :

- Customer acquisition source
- Customer demographics or preferences, where available
- Restaurant choices
- Customer complaints and reviews
- First-order experience in greater detail
- Promotional and campaign information

This could help identify what is actually different about customers who continue ordering and those who do not.

2. Cancellation Drivers

No strong standalone driver of Zomato cancellations was identified. Future analysis could examine :

- Cancellation reasons
- Restaurant-level operational events

- Rider availability
- Actual delivery delays
- Order acceptance and preparation events
- More detailed cancellation timestamps

A larger dataset with more cancellation cases would also make driver analysis more reliable.

3. Order-Ready Marking

Incorrect order-ready marking showed a clearer relationship with rider wait time, but the available data did not identify what causes incorrect or missed marking.

Further investigation could examine:

- Restaurant staff or outlet-level processes
- Order readiness timestamps
- Actual preparation completion time
- Peak-period workload
- Restaurant-specific operational practices

4. Customer Value

High-value customers generated a large share of revenue and ordered more frequently. Further analysis could examine what characteristics or experiences help customers become high-value customers and whether particular retention or engagement strategies increase their long-term value.

5. Restaurant Performance

Since most business activity is concentrated in Aura Pizzas and Swaad, future analysis could examine these restaurants in greater operational detail while also tracking whether smaller restaurants are growing over time.

Overall Limitation

The analysis provides evidence about what is happening in the business and which relationships are worth investigating, but it cannot fully explain why those patterns occur. Further investigation using richer data and controlled

experiments would be required to confirm the causes and measure the actual impact of potential business interventions.